

Project idea

Pick Up The Ball

Special Aspects of mobile autonomous Systems

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Agenda

- ❑ Idea
- ❑ Soft- and hardware used



Idea (1 / 2)

- Recognize an object and focus on a colour (initialize)
- Pick up the object
- Show the object

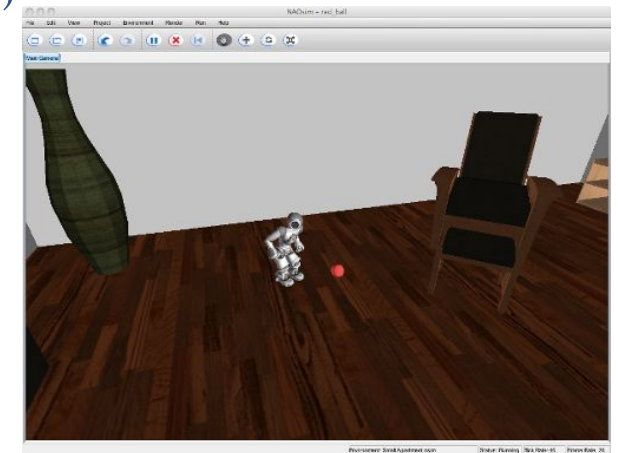
Difficulties: Recognizing the **ball**, **colour** and the **pick up** movement



Idea (2/2)

Development is divided into the following sequential tasks:

1. Simulate a NAO Robot and first experiment around using NAOsim
2. Try to recognize a single object (ball) using NAO in real-world
3. Try to recognize and calibrate the colour on this object
4. Follow the object with the head
5. Pick up the object using a *human motion*
6. Show the object



Soft- and hardware used

■ Hardware

- **NAO Robot**

- Humanoid robot
- Multiple cameras (*input, image sensors*)
- Various joints to emulate movements (*output, real-world feedback*)
- Programmable (*using cable, e.g.*)



■ Software

- Main Tool: **Jupyter Notebook**
- Simulation Tool: **NAOsim**
- Language: **Python 2.7.0** (+ *using NAOqi library as additional resource*)

Thank you for your attention!
Have any questions? - feel free to ask :)

